

DANIELS HARBOUR, NL, Canada

WMO: 711850

Lat: 50.2364N

Lon: 57.5811W

Elev: 62

StdP: 14.66

Time Zone: -3.50 (NAF)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-0.8	3.0	-10.6	3.1	0.3	-6.0	4.0	4.1	41.5	21.7	36.3	24.4	11.7	290	0.808

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	10.1	70.7	63.9	68.5	62.8	66.8	61.7	65.6	69.2	63.9	67.3	62.5	65.9	13.6	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.9	89.3	67.5	62.4	84.6	66.1	60.9	79.9	65.0	30.5	69.4	29.3	67.4	28.3	66.0	73.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 33.0	29.2	26.3	DB	-8.9	78.0	6.3	3.1	-13.4	80.2	-17.1	82.0	-20.6	83.7	-25.2	86.0	(4)
(5)			WB	-10.3	69.2	6.5	1.8	-15.0	70.5	-18.8	71.6	-22.5	72.6	-27.2	73.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	39.5	20.6	19.3	24.6	33.8	42.9	51.2	59.3	60.6	53.8	44.3	35.6	26.8		
	DBStd	15.78	8.82	9.27	9.02	6.53	5.62	5.80	4.68	4.42	6.08	6.07	7.36	7.89		
	HDD50	4716	911	861	788	487	230	53	2	1	29	198	437	719		
	HDD65	9317	1376	1281	1253	936	684	414	182	145	337	643	882	1184		
	CDD50	892	0	0	0	1	11	90	291	329	145	21	4	0		
	CDD65	17	0	0	0	0	0	0	6	8	3	0	0	0		
	CDH74	18	0	0	0	0	0	2	7	5	4	0	0	0		
	CDH80	1	0	0	0	0	0	0	0	1	0	0	0	0		
	Wind	WSAvg	12.7	14.5	13.5	13.1	12.9	11.3	11.5	11.2	12.8	12.7	13.6	14.2		
	Precipitation	PrecAvg	45.8	4.3	3.5	3.4	2.4	3.0	4.0	3.9	4.9	3.8	4.3	4.3		
		PrecMax	58.3	6.9	7.8	6.7	5.5	6.1	11.9	8.3	9.1	6.8	8.0	9.6		
		PrecMin	28.7	1.5	0.5	0.8	0.2	0.7	1.4	1.1	1.3	1.4	2.0	1.3		
		PrecStd	7.9	1.5	1.7	1.7	1.3	1.4	1.9	1.7	1.9	1.4	1.5	1.8		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	50.3	47.8	49.4	57.7	63.0	69.3	74.3	73.5	73.3	64.8	62.3	50.7		
		MCWB	47.9	44.3	43.4	50.4	54.0	60.8	65.6	64.9	66.0	56.3	55.9	48.1		
	2%	DB	40.5	39.4	42.8	51.6	58.1	65.4	70.3	70.6	67.9	59.1	55.6	43.9		
		MCWB	37.9	36.3	38.6	45.6	51.2	58.8	63.7	64.1	62.6	54.7	51.4	41.6		
	5%	DB	36.0	34.8	38.4	47.0	55.1	62.5	68.0	68.7	64.8	56.2	49.7	39.8		
		MCWB	33.8	32.0	34.8	41.8	49.1	57.1	62.3	63.3	60.5	52.3	46.7	37.7		
	10%	DB	32.7	32.0	35.3	42.5	51.9	60.0	65.9	67.2	62.6	53.3	45.4	36.8		
		MCWB	30.9	29.8	31.9	37.9	46.8	55.0	61.1	62.4	58.6	49.6	42.9	34.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	48.1	44.5	45.5	51.8	55.7	63.5	68.3	67.4	66.4	58.9	57.1	48.1		
		MCDB	50.7	47.3	49.2	56.1	60.4	67.0	71.1	71.0	72.2	63.1	61.3	50.6		
	2%	WB	38.3	36.6	39.5	45.8	52.7	60.4	65.3	65.6	63.5	55.4	51.8	42.1		
		MCDB	40.1	38.8	43.1	50.6	57.0	63.8	68.9	68.8	67.4	58.5	55.4	43.8		
	5%	WB	33.7	32.5	35.3	41.8	49.7	58.0	63.2	64.3	61.2	52.8	47.1	38.0		
		MCDB	35.4	34.2	38.2	46.2	53.8	61.7	66.6	67.6	64.2	55.8	49.5	39.4		
	10%	WB	30.4	29.9	32.4	38.5	46.9	55.8	61.8	63.0	59.1	50.1	43.2	34.7		
		MCDB	32.3	32.0	34.8	42.2	51.1	59.3	65.1	66.3	62.2	53.1	45.4	36.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	11.6	12.6	11.9	10.4	12.0	11.4	9.9	10.1	10.4	9.8	9.8	9.9		
		MCDBR	16.7	15.9	15.9	17.3	17.2	15.2	12.6	11.4	12.8	14.6	17.7	15.7		
		MCWBR	16.3	14.7	13.7	13.8	12.3	10.2	8.6	8.2	9.9	12.1	16.1	15.4		
	5% WB	MCDBR	16.5	16.9	15.9	16.9	16.6	13.6	11.4	10.1	12.1	13.8	17.2	15.6		
		MCWBR	15.8	15.8	14.3	13.9	12.6	9.9	8.6	8.1	10.2	12.2	16.0	15.5		
(40) Clear-Sky Solar Irradiance	taub		0.295	0.310	0.318	0.350	0.372	0.370	0.410	0.379	0.360	0.336	0.307	0.283		
	taud		2.293	2.240	2.281	2.329	2.374	2.399	2.345	2.430	2.462	2.485	2.467	2.393		
	Ebn at Noon		233	259	281	283	281	281	267	271	265	251	232	224		
	Edh at Noon		22	30	34	36	36	36	37	33	29	23	19	17		
(44) All-Sky Solar Radiation	RadAvg		243	454	824	1218	1552	1627	1612	1361	990	598	298	191		
	RadStd		28	60	100	146	125	167	138	88	88	49	30	19		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (13 neighbors)	N/A	N/A	+2.30	N/A	+0.68	+0.90	N/A	N/A	N/A	N/A

Nomenclature: See separate page